Generic Local Identifiability Does Not Imply Generic Global Identifiability in Linear Networks

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***Abstract*— We give a counterexample to the implication from generic local to generic global identifiability for linear networks with a fixed topology, independently unknown edge coefficients, and selected excited and measured nodes. A six-node network with nine unknown coefficients has exactly two compatible parameter vectors generically. Adding six excited and measured nodes preserves this ambiguity while making the added coefficients directly recoverable. The resulting twelve-node structure has forty-two independent unknown coefficients and zero diagonal. For every positive bound on the largest eigenvalue magnitude, a nonempty open set of parameters has two distinct compatible networks satisfying that bound. This holds for both real and complex coefficients. Other open regions have exactly one admissible solution. The proof uses a rational transformation exchanging the two solutions and a common limiting matrix whose fourth power vanishes. Exact rational-number certificates verify an explicit pair. The construction also extends algebraically when the unknown coordinates are whole rational transfer functions. Genericity in prescribed numerator and denominator coefficient families, and the requirement that every transfer function vanish at infinite frequency, remain separate questions.**

***Index Terms*— Identification, network analysis and control, linear systems.**

I. Introduction

A linear network relates node signals through coefficients on prescribed directed edges. Inputs excite selected nodes, and measurements record selected node signals. Network identification seeks to recover the unknown edge coefficients from the resulting input-output relation.[1]

Local identifiability means that the coefficients are unique among sufficiently close admissible choices. Global identifiability requires uniqueness among all admissible choices. For scalar coefficients, a property is generic if it holds outside a set of zero volume. A derivative test for local uniqueness can leave a second, isolated explanation of the same input-output relation.

Legat and Hendrickx developed a generic local identifiability test for partial excitation and measurement and asked whether local and global identifiability can differ in this setting [2]. The two notions coincide when all nodes are excited or all nodes are measured [3]. A further equivalence result holds for a specified class of two-part networks with known internal coefficients [4]. The general question remains open in the discussion of [3, Sec. II-C] and [5, Sec. 12.2]. The latter reports an unsuccessful search for a counterexample with up to five unknown edges, subject to specified structural restrictions.

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[1]Drafting assistance: OpenAI Codex [1].

This letter answers the question for the cited scalar formulation. The topology and node selections are fixed, every permitted edge has an independent unknown coefficient, and excitation and measurement are both partial. A six-node construction has exactly two compatible vectors generically. Six added excited and measured nodes preserve the ambiguity and make both spectral radii, the largest eigenvalue magnitudes, arbitrarily small. The resulting structure has forty-two independent coefficients. For every positive spectral-radius bound, ambiguity occurs on open subsets of both the full complex and real parameter spaces. Both alternatives satisfy the same bound, including the restriction in [3]. The node and edge counts are constructive upper bounds; minimality remains open.

The scalar setting allows direct feedthrough, meaning an instantaneous dependence between connected node signals. Constant real coefficients therefore also give transfer-function examples. We further use the algebraic framework of [6], where each coordinate is a whole rational transfer function and genericity excludes polynomial equations in these functions. This differs from varying the numerical numerator and denominator coefficients at a prescribed dynamic order. Such coefficient families, and strictly proper functions that vanish at infinite frequency, require separate analysis.

Section II states the admissible classes and the precise identifiability definitions. Sections III and IV develop the two constructions. Section V proves the spectral-radius result and gives an explicit pair with rational entries. Section VI extends the result to rational transfer functions under an algebraic definition of generic identifiability.

II. Network Description and Identifiability

A transfer function describes a linear relation between signals in the frequency domain. We follow the scalar formulation of [2, Secs. I and III] and [3, Sec. III-B]: each edge coefficient represents a transfer-function value at a fixed frequency and can therefore be complex.[2]

Let n be the number of nodes, numbered $1, \dots, n$, and let E be a prescribed set of ordered pairs (i, j), where $i, j \in \{1, \dots, n\}$ and $i \neq j$. An edge (i, j) carries the signal from node j to node i. Write $m = |E|$, fix an ordering of these edges, and collect their coefficients in a vector $\theta \in \mathbb{C}^m$. The network matrix $G(\theta) \in \mathbb{C}^{n \times n}$ places these coefficients in the positions in E and has zeros elsewhere. Each edge has

[2]Drafting assistance: OpenAI Codex [1].

its own independent parameter. In particular, $G(\theta)$ has zero diagonal.

Let n_B and n_C be the numbers of excited and measured nodes. The fixed matrix $B \in \{0,1\}^{n\times n_B}$ has distinct columns of the identity matrix, and $C \in \{0,1\}^{n_C\times n}$ has distinct rows of the identity matrix. Thus B inserts inputs at the excited nodes and C selects the measured signals. At the chosen frequency, let $w \in \mathbb{C}^n$, $r \in \mathbb{C}^{n_B}$, and $y \in \mathbb{C}^{n_C}$ be the complex amplitudes of the node signals, inputs, and measurements, respectively. The network equations are

$$w = G(\theta)w + Br, \qquad y = Cw. \tag{1}$$

For a positive integer d, let I_d denote the $d \times d$ identity matrix. We call (1) well posed if $I_n - G(\theta)$ is invertible. In that case the transfer matrix from the external inputs r to the measurements y, evaluated at the chosen frequency, is

$$\Phi(\theta) = C(I_n - G(\theta))^{-1}B. \tag{2}$$

Identification concerns recovery of θ from $\Phi(\theta)$. The fiber at a transfer matrix is the set of admissible parameter vectors producing it.

Define the set of well-posed parameters by

$$\Omega = \{\theta \in \mathbb{C}^m : \det(I_n - G(\theta)) \neq 0\}.$$

For a matrix, let ρ denote its spectral radius. For $0 < \varepsilon \leq 1$, define the more restricted set

$$\Omega_\varepsilon = \{\theta \in \mathbb{C}^m : \rho(G(\theta)) < \varepsilon\}. \tag{3}$$

The strict inequality implies $\Omega_\varepsilon \subseteq \Omega$. Openness and volume are understood by identifying $\mathbb{C}^m$ with $\mathbb{R}^{2m}$ through real and imaginary parts. Both parameter sets are open in this topology.

Let $\mathcal{D}$ be either Ω or Ω_ε. The network is *locally identifiable at* $\theta \in \mathcal{D}$ if some neighborhood N of θ satisfies the following implication for every alternative vector $\widetilde{\theta}$:

$$\widetilde{\theta} \in N \cap \mathcal{D}, \quad \Phi(\widetilde{\theta}) = \Phi(\theta) \quad \Longrightarrow \quad \widetilde{\theta} = \theta.$$

It is *globally identifiable at* θ *in* $\mathcal{D}$ if the same implication holds for every $\widetilde{\theta} \in \mathcal{D}$. Thus any competing network must satisfy the same admissibility restriction.

The structure is generically locally or globally identifiable in $\mathcal{D}$ if the corresponding property holds outside a subset of $\mathcal{D}$ of Lebesgue measure zero, that is, zero volume. We use the complex Jacobian, the matrix of derivatives of the entries of Φ with respect to the complex coordinates of θ. All ranks are over $\mathbb{C}$. We prove generic local identifiability by showing that this Jacobian has column rank m except where a nonzero polynomial vanishes. Such an exceptional set has zero volume in $\mathbb{R}^{2m}$. For the rational map Φ, complex rank m gives real rank $2m$ after separating real and imaginary parts. Applying the inverse function theorem to m suitable complex entries of Φ then gives local uniqueness. Conversely, a nonempty open set of parameters with distinct admissible alternatives suffices to disprove generic global identifiability. This conclusion alone does not say that almost every admissible parameter vector has an alternative.

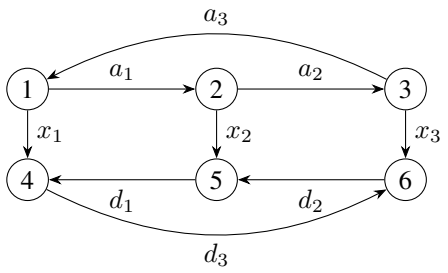


Fig. 1. The six-node construction. Nodes $1, 2, 3$ are excited; nodes $4, 5, 6$ are measured. Each labeled edge has an independent unknown coefficient.

The generic claims in Sections III–V refer to the independent scalar parameters. Section VI considers independent rational transfer functions as the unknowns.

III. A Six-Node Ambiguity

Consider nine independent complex parameters a_i, d_i, x_i, indexed by $i \in \{1, 2, 3\}$. In this section and in formulas using these parameters, subscripts are cyclic: $i+1 = 1$ when $i = 3$, and $i - 1 = 3$ when $i = 1$.[3] Let diag form a diagonal matrix from its arguments, and define

$$A = \begin{pmatrix} 0 & 0 & a_3 \\ a_1 & 0 & 0 \\ 0 & a_2 & 0 \end{pmatrix}, \qquad D = \begin{pmatrix} 0 & d_1 & 0 \\ 0 & 0 & d_2 \\ d_3 & 0 & 0 \end{pmatrix}, \qquad X = \mathrm{diag}(x_1, x_2, x_3), \tag{4}$$

In block matrices, zero blocks have the dimensions determined by their positions. Set

$$G_b = \begin{pmatrix} A & 0 \\ X & D \end{pmatrix}, \quad B_b = \begin{pmatrix} I_3 \\ 0 \end{pmatrix}, \quad C_b = \begin{pmatrix} 0 & I_3 \end{pmatrix}. \tag{5}$$

This is (1) with six nodes, the first three excited and the last three measured. Figure 1 shows the edges. The within-part matrices A, D are unknown. In the separable networks of [4], these matrices are known and only the connections between parts are unknown; that equivalence theorem therefore does not apply here. Define the products $\alpha = a_1a_2a_3$ and $\delta = d_1d_2d_3$. Then

$$\det(I_6 - G_b) = (1 - \alpha)(1 - \delta).$$

For well-posed parameters, denote the transfer matrix by

$$F = C_b(I_6 - G_b)^{-1}B_b = (I_3 - D)^{-1}X(I_3 - A)^{-1}. \tag{6}$$

For nonzero x_i, define the auxiliary quantities

$$k_i = a_id_i\frac{x_{i+1}}{x_i}, \qquad h_i = 1 + k_i + k_ik_{i+1}. \tag{7}$$

Let $\mathcal{D}_b \subset \mathbb{C}^9$ be the domain specified by

$$\begin{aligned} x_1x_2x_3 &\neq 0, \\ \alpha\delta(1 - \alpha)(1 - \delta)(1 - \alpha\delta)h_1h_2h_3 &\neq 0. \end{aligned} \tag{8}$$

Define T_b on the nine parameters by

$$\eta_i = \frac{k_ih_{i+1}}{h_i}, \qquad (a_i', d_i', x_i') = \frac{1}{\eta_i}(a_i, d_i, x_i). \tag{9}$$

[3]Drafting assistance: OpenAI Codex [1].

Primes denote the resulting alternative values.

Proposition 1: The structure (5) is generically locally identifiable in Ω. For every parameter vector in $\mathcal{D}_b$, exactly two well-posed vectors give its transfer matrix F, and both are locally identifiable. The map T_b preserves $\mathcal{D}_b$ and is a rational involution: it has rational coordinates and applying it twice returns the original vector.

Proof: Assume first that $x_1x_2x_3 \neq 0$, so that F is invertible. Introduce coordinates $u_i = a_i/x_i$, $v_i = d_i/x_i$, and $z_i = 1/x_i$. This change is invertible, with $a_i = u_i/z_i$, $d_i = v_i/z_i$, and $x_i = 1/z_i$. The inverse transfer matrix is

$$\begin{aligned} F^{-1} &= (I_3 - A)X^{-1}(I_3 - D) \\ &= \begin{pmatrix} z_1 + u_3v_3/z_3 & -v_1 & -u_3 \\ -u_1 & z_2 + u_1v_1/z_1 & -v_2 \\ -v_3 & -u_2 & z_3 + u_2v_2/z_2 \end{pmatrix}. \end{aligned} \tag{10}$$

Thus the off-diagonal entries determine u_i, v_i. Let $e_i = (F^{-1})_{ii}$ be the diagonal entries. They satisfy

$$e_i = z_i + \frac{u_{i-1}v_{i-1}}{z_{i-1}}. \tag{11}$$

With u_i, v_i fixed, the Jacobian of these three equations with respect to (z_1, z_2, z_3), denoted by J_z, has determinant

$$\det J_z = 1 - \frac{u_1v_1u_2v_2u_3v_3}{z_1^2z_2^2z_3^2} = 1 - \alpha\delta. \tag{12}$$

The six directly determined coordinates and (12) give Jacobian rank nine when $\alpha\delta \neq 1$. Inverting an invertible transfer matrix preserves rank. This proves generic local identifiability.

Expanding h_i and using $k_1k_2k_3 = \alpha\delta$ gives

$$\begin{aligned} \eta_i - 1 &= \frac{\alpha\delta - 1}{h_i}, \\ \eta_i + \frac{k_{i-1}}{\eta_{i-1}} &= 1 + k_{i-1}, \qquad \eta_1\eta_2\eta_3 = \alpha\delta. \end{aligned} \tag{13}$$

The transformation fixes u_i, v_i and gives $z_i' = \eta_i z_i$. Since $u_{i-1}v_{i-1}/z_{i-1} = k_{i-1}z_i$, the second identity in (13) preserves (11), and hence F. The first identity makes the alternatives distinct when $\alpha\delta \neq 1$. Denote the products of the primed cycle coefficients by $\alpha' = \prod_{i=1}^3 a_i'$ and $\delta' = \prod_{i=1}^3 d_i'$. The last identity gives

$$\alpha' = \frac{1}{\delta}, \qquad \delta' = \frac{1}{\alpha}. \tag{14}$$

Thus the alternative is well posed on $\mathcal{D}_b$, and $1 - \alpha'\delta' = 1 - 1/(\alpha\delta) \neq 0$ gives rank nine there. Applying (7) and (9) to the primed parameters also gives

$$h_i' = \frac{h_{i+1}}{k_{i+1}k_{i+2}}, \qquad \eta_i' = \eta_i^{-1}. \tag{15}$$

All factors excluded in (8) remain nonzero. Hence $T_b(\mathcal{D}_b) = \mathcal{D}_b$ and T_b^2 is the identity.

It remains to bound the number of solutions. Define $b_i = u_iv_i$. Equation (11) gives successively

$$z_2 = e_2 - b_1/z_1, \quad z_3 = e_3 - b_2/z_2, \quad z_1 = e_1 - b_3/z_3.$$

Substitution and multiplication by the denominators yield

$$\begin{aligned} 0 = {} & (b_2 - e_2e_3)z_1^2 \\ & + (b_1e_3 - b_2e_1 - b_3e_2 + e_1e_2e_3)z_1 \\ & + b_1b_3 - b_1e_1e_3. \end{aligned} \tag{16}$$

Its leading coefficient is $b_2 - e_2e_3 = -z_2z_3h_1$, which is nonzero on $\mathcal{D}_b$. There are therefore at most two values of z_1. Each determines z_2, z_3 and all original parameters. A candidate with some $x_i = 0$ cannot yield an invertible F, by (6). Formula (9) supplies the distinct second solution. After clearing denominators, the exclusions in (8) are zeros of nonzero polynomials: $a_i = d_i = -1/2$, $x_i = 1$ satisfy all the inequalities. Thus $\mathcal{D}_b$ has full volume in Ω, completing the generic claim. ∎

The six-node construction alone does not give two alternatives in Ω_1. Indeed, its block triangular form gives

$$\rho(G_b) = \max\{|\alpha|^{1/3}, |\delta|^{1/3}\}.$$

If both products have magnitude below one, (14) makes the alternative violate the same restriction. The next construction changes the full network eigenvalues while retaining the ambiguity.

IV. Adding Excited and Measured Nodes

Let $w_b \in \mathbb{C}^6$ collect the original node amplitudes. Add six nodes with amplitudes $w_c \in \mathbb{C}^6$, all excited and measured. Their purpose is to change the complete network spectrum while preserving the two base solutions.[4] Let $r_b \in \mathbb{C}^3$ and $s \in \mathbb{C}^6$ be the inputs at the original and added nodes, and $y_b = C_bw_b$ the original measurements. Let $K \in \mathbb{C}^{3\times6}$, $L_c \in \mathbb{C}^{6\times3}$, and $J \in \mathbb{C}^{6\times6}$ specify the additional connections. Define the augmented network and its selections by

$$\begin{aligned} \mathcal{G} &= \begin{pmatrix} G_b & B_bK \\ L_cC_b & J \end{pmatrix}, \\ \mathcal{B} &= \begin{pmatrix} B_b & 0 \\ 0 & I_6 \end{pmatrix}, \qquad \mathcal{C} = \begin{pmatrix} C_b & 0 \\ 0 & I_6 \end{pmatrix}. \end{aligned} \tag{17}$$

This again has the form (1); its twelve node amplitudes are the six original amplitudes w_b followed by w_c. The matrices K, L_c, J connect the added nodes to the original excited and measured nodes and to each other. Denote its transfer matrix by

$$H = \mathcal{C}(I_{12} - \mathcal{G})^{-1}\mathcal{B}. \tag{18}$$

Figure 2 groups the nodes and added connections.

Lemma 1: Suppose $I_6 - G_b$, X, and $I_{12} - \mathcal{G}$ are invertible. Then H is invertible and

$$H^{-1} = \begin{pmatrix} F^{-1} & -K \\ -L_c & I_6 - J \end{pmatrix}. \tag{19}$$

Consequently H determines each entry of K, L_c, J, and, on this domain, equality of augmented transfer matrices is equivalent to equality of F, K, L_c, J. Every well-posed candidate with this H also has invertible X and $I_6 - G_b$. If the original base parameters belong to $\mathcal{D}_b$, the augmented

[4] Drafting assistance: OpenAI Codex [1].

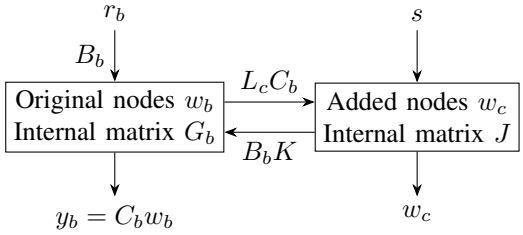


Fig. 2. Augmentation of the six-node network. Each box groups six node equations. Every added node has its own input and measurement; the horizontal arrows specify the connections between the groups.

fiber consists of exactly the two base solutions with the same K, L_c, J.

Proof: The node equations give

$$y_b = F(r_b + Kw_c), \qquad w_c = L_cy_b + Jw_c + s.$$

Thus

$$\begin{pmatrix} r_b \\ s \end{pmatrix} = \begin{pmatrix} F^{-1} & -K \\ -L_c & I_6 - J \end{pmatrix} \begin{pmatrix} y_b \\ w_c \end{pmatrix}.$$

By well-posedness, (y_b, w_c) is obtained from (r_b, s) through the square matrix H. The displayed matrix is therefore a left inverse of H, proving (19).

For any well-posed candidate producing this H, suppose $X\xi = 0$ for some nonzero $\xi \in \mathbb{C}^3$. The node amplitudes $(\xi, 0, 0)$, in groups of sizes three, three, and six, give zero output for input $((I_3 - A)\xi, 0)$. Well-posedness makes this input nonzero, contradicting invertibility of H. Thus X is invertible. Eliminating the first three node amplitudes makes the upper left block of H^{-1} equal to $(I_3 - A)X^{-1}(I_3 - D)$. That block is the original invertible F^{-1}, so $I_3 - A$ and $I_3 - D$ are also invertible. Every candidate therefore satisfies the lemma's assumptions.

Proposition 1 leaves exactly two base candidates on $\mathcal{D}_b$. For either, (19) determines (y_b, w_c) from the inputs, and $X^{-1}(I_3 - D)y_b$ gives the first three node amplitudes. Both augmented candidates are therefore well posed. ∎

This identity also preserves the rank information needed for local identifiability. If q_c entries of K, L_c, J are independently unknown and the remaining entries are fixed zeros, their derivatives occupy positions in (19) disjoint from those of F^{-1}. The augmented Jacobian rank is therefore $q_c + 9$ wherever the base Jacobian has rank nine. The choice below makes both compatible augmented matrices satisfy the spectral-radius restriction. Moreover, (13) shows that T_b changes every base coefficient on $\mathcal{D}_b$. Knowing any one of these coefficients selects one of the two solutions, for both constructions.

V. Ambiguity With Arbitrarily Small Spectral Radius

Use superscript T for transpose. Define the cyclic permutation matrix U, and then V, by

$$U = \begin{pmatrix} 0 & 0 & 1 \\ 1 & 0 & 0 \\ 0 & 1 & 0 \end{pmatrix}, \qquad V = U + U^T.$$

These matrices satisfy $U^3 = I_3$ and $U^T = U^2$.[5] For real $t > 0$ and $x \neq 0$, take $A = -tU,\ D = -tU^T,\ X = xI_3$. Write $F(t, x)$ for the corresponding transfer matrix (6). Then

$$F(t, x)^{-1} = x^{-1}\big[(1 + t^2)I_3 + tV\big], \tag{20}$$

so $F(t, 1) = F(t^{-1}, t^{-2})$. The two base matrices coincide at $t = 1$; denote this common matrix by G_{b0}.

We design the added connections at this common point. Seek K_0, L_0 such that $E_+ = G_{b0} + B_bK_0$ and $E_- = G_{b0} - L_0C_b$ have zero squares, and set

$$J_0 = G_{b0} + B_bK_0 - L_0C_b.$$

This choice makes the equations in coordinates $(w_b, w_b - w_c)$ block upper triangular with diagonal blocks E_+, E_-. Their zero squares make the complete matrix's fourth power zero. Using $U^3 = I_3$, suitable choices are

$$K_0 = \begin{pmatrix} V & -U \end{pmatrix}, \quad L_0 = \begin{pmatrix} U^T \\ -V \end{pmatrix}, \quad J_0 = \begin{pmatrix} U^T & -V \\ I_3 & U \end{pmatrix}. \tag{21}$$

Combining them with the varying base gives

$$M(t, x) = \begin{pmatrix} -tU & 0 & V & -U \\ xI_3 & -tU^T & 0 & 0 \\ 0 & U^T & U^T & -V \\ 0 & -V & I_3 & U \end{pmatrix}. \tag{22}$$

All blocks in this display have size 3×3. Use the nonzero positions in $M(t, x)$ as the permitted edge set for $\mathcal{G}$. This pattern does not depend on the chosen nonzero values of t, x. Assign a separate independent unknown coefficient to every permitted position. There are nine in G_b, nine in K, nine in L_c, and fifteen in J, giving $m = 42$. Thus the repeated numerical values in (22) specify particular parameter choices in a class with forty-two independent complex coefficients.

The selections $\mathcal{B}, \mathcal{C}$ in (17) excite nodes 1–3 and 7–12 and measure nodes 4–12. There are no diagonal edges. Let $\theta \in \mathbb{C}^{42}$ collect the independent coefficients of this structure, and apply (3) to $\mathcal{G}(\theta)$.

Theorem 1: For every $0 < \varepsilon \leq 1$, this twelve-node structure is generically locally identifiable in Ω_ε. There is a nonempty open set $W_\varepsilon \subset \Omega_\varepsilon$ such that every $\theta \in W_\varepsilon$ has a distinct alternative $\widetilde{\theta} \in \Omega_\varepsilon$ with the same transfer matrix H. Both alternatives are locally identifiable. Consequently, the structure is not generically globally identifiable in Ω_ε. The same conclusions hold on the real parameter domain $\Omega_\varepsilon \cap \mathbb{R}^{42}$.

Proof: We first give equal transfer matrices, then make both spectral radii small, and finally establish ambiguity on an open set of all forty-two parameters.

Equal transfer matrices: The two augmented matrices

$$\mathcal{G}_1(t) = M(t, 1), \qquad \mathcal{G}_2(t) = M(t^{-1}, t^{-2}) \tag{23}$$

use the same K_0, L_0, J_0. By Lemma 1, their transfer matrices agree whenever both augmented networks are well posed. They are distinct for $t \neq 1$.

[5]Drafting assistance: OpenAI Codex [1].

Small spectral radii: At $t = 1$, both matrices in (23) equal $M(1,1)$. The design above gives

$$\begin{aligned} E_+ &= G_{b0} + B_b K_0 = \begin{pmatrix} U^T & -U \\ I_3 & -U^T \end{pmatrix}, \\ E_- &= G_{b0} - L_0 C_b = \begin{pmatrix} -U & -U^T \\ I_3 & U \end{pmatrix}. \end{aligned} \tag{24}$$

Changing coordinates from (w_b, w_c) to $(w_b, w_b - w_c)$ makes $M(1,1)$ similar to the block upper triangular matrix

$$\begin{pmatrix} E_+ & -B_b K_0 \\ 0 & E_- \end{pmatrix}. \tag{25}$$

The identities for U imply $E_+^2 = E_-^2 = 0$. Squaring (25) therefore gives a matrix whose only possibly nonzero block is the upper right block; its square is zero. Consequently

$$M(1,1)^4 = 0. \tag{26}$$

For a $d \times d$ matrix Q, let $\|Q\|_\infty = \max_{1 \le i \le d} \sum_{j=1}^{d} |Q_{ij}|$, its maximum absolute row sum. The standard eigenvalue bound gives

$$\rho(Q)^4 = \rho(Q^4) \le \|Q^4\|_\infty. \tag{27}$$

Both matrices in (23) converge to $M(1,1)$ as $t \to 1$. Their fourth-power norms consequently tend to zero. For any prescribed $0 < \varepsilon \le 1$, choose $t \ne 1$ sufficiently close to one that both norms are below ε^4. Equation (27) then places both distinct matrices in Ω_ε, so their transfer matrices are well defined and equal.

Local identifiability: The thirty-three independent added coefficients are determined directly in (19). The base Jacobian has rank nine by (12), since $\alpha\delta = t^6$ at the first point of (23) and $\alpha\delta = t^{-6}$ at the second. Both augmented Jacobians therefore have rank $33+9 = 42$. All their entries are rational functions of the forty-two parameters. A minor, meaning the determinant of a square submatrix, that is nonzero at one of these points has a numerator that is a nonzero polynomial. Outside its zero set, the Jacobian of H has full column rank. This establishes generic local identifiability in Ω_ε.

An open set of ambiguous parameters: Extend T_b to an involution $\mathcal{T}$ on the forty-two parameters by leaving the thirty-three added coefficients unchanged. At the first point of (23), one has $k_i = t^2$, $h_i = 1 + t^2 + t^4$, and $\eta_i = t^2$; thus $\mathcal{T}$ maps it to the second point. All denominators are nonzero there. The transformation is rational and continuous in a neighborhood, and preserves H by Lemma 1.

At these two points the fourth-power bounds are strict, the alternatives are distinct, and both Jacobians have full column rank. These properties persist in a sufficiently small open neighborhood W_ε of the first point and its image under $\mathcal{T}$. Shrink the neighborhood further, if necessary, so that the base inverses in the lemma exist and all permitted coefficients remain nonzero. Then every $\theta \in W_\varepsilon$ and its alternative $\mathcal{T}(\theta)$ belong to Ω_ε, have equal transfer matrices, and are distinct and locally identifiable. The neighborhood is open in $\mathbb{C}^{42}$, so it has positive volume in the real and imaginary coordinates of $\mathbb{R}^{84}$. This disproves generic global identifiability. The pair is real and $\mathcal{T}$ preserves real coefficients. The neighborhood contains a real open set, and the same nonzero Jacobian minor establishes generic local identifiability there. ∎

A. An explicit pair with rational constant coefficients

For $\varepsilon = 1$, choose $t = 99/100$ in (23). The common inverse transfer matrix, obtained from (19), is

$$H^{-1} = \begin{pmatrix} (1+t^2)I_3 + tV & -V & U \\ -U^T & I_3 - U^T & V \\ V & -I_3 & I_3 - U \end{pmatrix}. \tag{28}$$

Exact multiplication of the rational matrices gives

$$\begin{aligned} \|\mathcal{G}_1 (99/100)^4\|_\infty &= \frac{33910001}{100000000} < 1, \\ \|\mathcal{G}_2 (99/100)^4\|_\infty &= \frac{16976126}{32019867} < 1. \end{aligned} \tag{29}$$

These inequalities certify admissibility through (27). Their spectral radii are approximately 0.509563 and 0.404210, respectively. The exact equalities (28) and (29) make the pair directly checkable with rational arithmetic. Theorem 1 establishes the open-set conclusion independently of these numerical values.

B. Admissible solution counts and conditioning

There are also open subsets of Ω_ε on which the network is globally identifiable relative to that domain. To see this, choose $0 < t < \varepsilon$ with $x = 1$, initially setting $K = L_c = J = 0$. The full network and its alternative have spectral radii t and t^{-1}, respectively. Both are well posed and the base parameters lie in $\mathcal{D}_b$. Lemma 1 gives exactly these two candidates. Small independent perturbations, with all permitted added entries nonzero, keep the first radius below ε and the second above it. Thus one admissible solution persists on an open set, alongside the two-solution regions of Theorem 1. Failure of generic global identifiability therefore does not imply that almost every admissible network has an admissible alternative.

The spectral-radius bound concerns eigenvalues. At the limit, $\rho(M(1,1)) = 0$ while $\|M(1,1)\|_\infty = 4$. Along $M(t,1)$, the reduced Jacobian in (12) is $J_z = I_3 - t^2 U$. Writing $\sigma_{\min}$ for the smallest singular value, the cyclic permutation gives

$$\sigma_{\min}(J_z) = |1 - t^2| \longrightarrow 0 \quad (t \to 1).$$

Its inverse therefore amplifies some perturbations by $|1 - t^2|^{-1}$. The theorem supplies no uniform sensitivity bound or neighborhood size as ε decreases.

VI. Transfer Functions and Conclusions

A. Algebraic identifiability over rational functions

Let λ be a symbolic complex frequency variable and let $\mathbb{K} = \mathbb{C}(\lambda)$ be the field of ratios of polynomials in λ with complex coefficients. We now replace the scalar edge coefficients by elements of $\mathbb{K}$, keeping the same network structures and formulas for the transfer matrices. Each coordinate of $\theta \in \mathbb{K}^m$ is then a rational function of λ. Equality of transfer matrices means an identity of rational functions.[6]

[6]Drafting assistance: OpenAI Codex [1].

We use the algebraic framework of [6, Sec. V]. For the algebraic calculation, the edge coordinates are independent variables over $\mathbb{K}$. Solutions are counted in an algebraic closure $\overline{\mathbb{K}}$, a field containing $\mathbb{K}$ in which every nonconstant polynomial has a root. An algebraic set is a set defined by polynomial equations in these coordinates. An algebraically generic property holds outside a proper algebraic subset, meaning one smaller than the full parameter domain or the image of the map $\theta \mapsto \Phi(\theta)$, as appropriate. Generic local identifiability in the algebraic sense means that the generic fiber has dimension zero, equivalently that it consists of finitely many isolated points. Generic global identifiability requires one point. Admissibility here excludes zeros of the determinants and denominators required by the network equations.

In the notation of [6], the matrices P, Q_r, R_y, S_{yr} define the closed-loop transfer function $R_y P^{-1} Q_r + S_{yr}$ from external inputs to measured outputs. Our augmented network is its specialization

$$P = I_{12} - \mathcal{G}, \qquad Q_r = \mathcal{B}, \qquad R_y = \mathcal{C}, \qquad S_{yr} = 0.$$

Proposition 2: Over $\mathbb{K}$, the six-node structure (5) and the twelve-node structure of Theorem 1 each have exactly two points in the generic algebraic fiber. Both are therefore generically locally identifiable in the algebraic sense and fail generic global identifiability. If one generic parameter vector has coordinates in $\mathbb{K}$, its distinct alternative also has coordinates in $\mathbb{K}$.

Proof: The identities and solution counts in Proposition 1 and Lemma 1 use only field operations and apply over $\overline{\mathbb{K}}$. They give exactly two solutions on the nonempty domain (8), with well-posedness also imposed for the augmented network. Formula (9) is rational, so it preserves $\mathbb{K}$-valued coordinates; the added coefficients remain unchanged. ■

B. Proper constant networks

A rational transfer function is *proper* if its numerator degree does not exceed its denominator degree, and *strictly proper* if the inequality is strict. Each real coefficient in (23) defines a constant proper rational transfer function. The matrix $(I_{12} - \mathcal{G})^{-1}$, which maps additive inputs at all nodes to the node signals, is also constant. It maps bounded inputs to bounded signals, so the network is stable in this input-output sense. Both networks have identical transfer matrices and satisfy the spectral-radius restriction at every frequency.

Algebraically generic uniqueness also fails after imposing properness, stability, and the spectral-radius bound. Indeed, $W_\varepsilon \cap \mathbb{R}^{42}$ from Theorem 1 contains a nonempty real open set of ambiguous constant networks. This set is Zariski dense over $\overline{\mathbb{K}}$, meaning that no nonzero polynomial in the forty-two edge coordinates vanishes on all of it. To see this, take a real open box contained in the set. Fix all but one coordinate in their respective intervals. A polynomial vanishing throughout the box has infinitely many roots in the remaining coordinate, so each of its coefficients is zero. Repeating this argument for the other coordinates proves that the polynomial is zero. Thus no proper algebraic exceptional set can contain all these admissible ambiguous networks. Their transfer matrices are also Zariski dense in the image of the map $\theta \mapsto H(\theta)$. Indeed, suppose a polynomial vanishes on all these transfer matrices but not on the full image. Substituting $H(\theta)$ and clearing denominators gives a nonzero polynomial that vanishes on the real open set, a contradiction.

This density statement uses whole transfer functions as coordinates. In a family parameterized by numerator and denominator coefficients at a prescribed positive order, constants may be excluded or form a lower-dimensional subset. The parameter spaces differ. Preservation of properness, stability, or a prescribed order by $\mathcal{T}$ for arbitrary nonconstant parameters requires a separate argument. The strictly proper case also remains open.

C. Reproducibility and conclusions

The companion verification archive [7] contains code, dependencies, and exact outputs. It checks the symbolic involution, nonsymmetric examples, equality of the transfer matrices, and the power bounds. The Jacobian checks use all forty-two independent edge positions. It also verifies an explicit example with only one spectrally admissible solution. The constructions show that derivative rank tests establish local recovery without ensuring global uniqueness for this network class, even under a common spectral-radius bound.

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